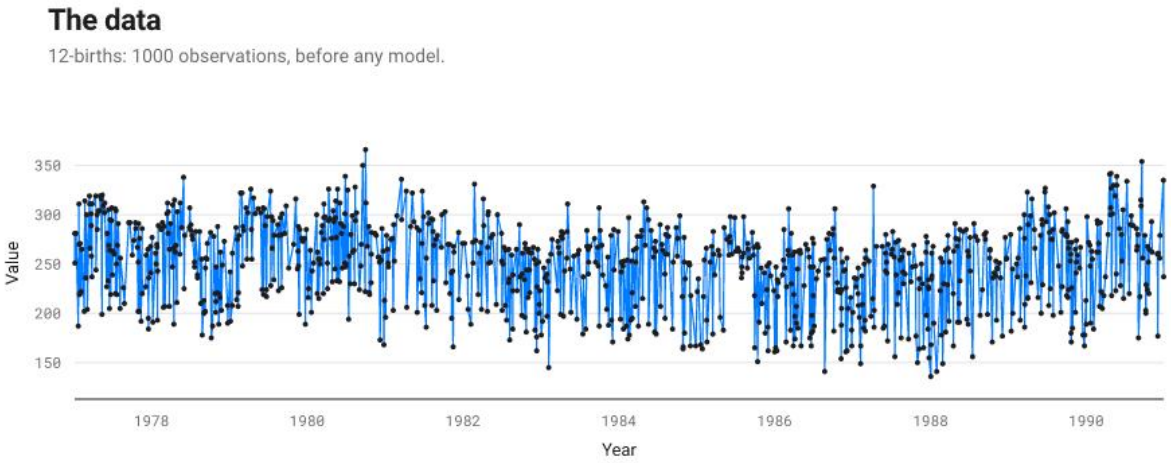
An automatic report for the dataset: 12-births

Some Math empowered by some AI guided by some researchers

10 November 2026

Produced by ***autostat****, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

The structure search selected `WN + Per + Per + SE` (Bayesian information criterion 9398.3; negative log likelihood 4668.0 on 1000 points). It decomposes into 4 additive components, listed by how much each reduces extrapolation error:

1. A very smooth function.
2. A periodic function with a period of 7.0 days.
3. A periodic function with a period of 6.0 months.
4. Uncorrelated noise.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			249	
1	9.7	9.7	33.5	86.5
2	67.1	63.5	19.1	43.1
3	68.7	4.9	18.7	2.1
4	100.0	100.0	18.7	0.0

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$y(t) = f_1(t) + f_2(t) + f_3(t) + f_4(t)$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}\left(0,\ \ 66779\,\exp\left(-\frac{(t-t')^2}{2\cdot 10.87^2}\right)\right)$$

$$f_2(t) \sim \mathrm{GP}\left(0,\ \ 825536\,\frac{\exp\left(1.135\cos\frac{2\pi(t-t')}{0.01917}\right) - I_0(1.135)}{e^{1.135} - I_0(1.135)}\right)$$

$$f_3(t) \sim \mathrm{GP}\left(0,\ \ 32.94\,\frac{\exp\left(0.1694\cos\frac{2\pi(t-t')}{0.4963}\right) - I_0(0.1694)}{e^{0.1694} - I_0(0.1694)}\right)$$

$$f_4(t) \sim \mathrm{GP}(0,\ \ 687.1\,\delta_{tt'})$$

These components have a closed form, read from their posterior mean:

$$\begin{aligned} f_1(t) &\approx -\left(7.157u - 29.86\right)\sin\left(0.2388u\right) + 258.9, \quad u \\ &= t - 1977 \end{aligned}$$

$$\begin{aligned} f_2(t) \approx\ & 38.53\cos\left(\frac{2\pi\cdot(t-1977)}{0.01917} + 1.22\right) + 19.15\cos\left(\frac{2\pi\cdot 2\cdot(t-1977)}{0.01917} - 0.6282\right) \\ & + 2.626\cos\left(\frac{2\pi\cdot 3\cdot(t-1977)}{0.01917} - 1.806\right) + 3.398\cos\left(\frac{2\pi\cdot 4\cdot(t-1977)}{0.01917} - 0.7962\right) \\ & + 5.707\cos\left(\frac{2\pi\cdot 5\cdot(t-1977)}{0.01917} - 2.406\right) + 5.111\cos\left(\frac{2\pi\cdot 6\cdot(t-1977)}{0.01917} + 2.486\right) \\ & + 3.098\cos\left(\frac{2\pi\cdot 7\cdot(t-1977)}{0.01917} + 0.1273\right) + 0.7295\cos\left(\frac{2\pi\cdot 8\cdot(t-1977)}{0.01917} + 1.556\right) \end{aligned}$$

$$\begin{aligned} f_3(t) \approx\ & 7.008\cos\left(\frac{2\pi\cdot(t-1977)}{0.4963} + 2.514\right) + 0.948\cos\left(\frac{2\pi\cdot 2\cdot(t-1977)}{0.4963} - 0.751\right) \\ & + 0.02429\cos\left(\frac{2\pi\cdot 3\cdot(t-1977)}{0.4963} + 0.7833\right) \end{aligned}$$

A periodic kernel is exactly a sum of cosines, so a periodic component is a Fourier series in its period; it is written up to the harmonic after which the rest weighs less than 0.1

The other components are smooth, quasi-periodic or noise without a short formula: a draw from such a process has no finite closed form; its covariance above is the formula.

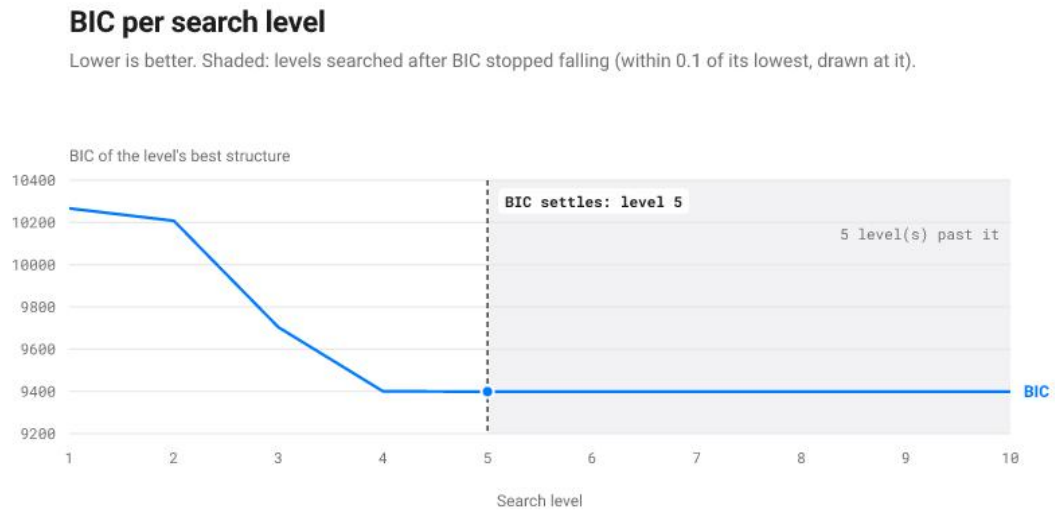
Closed form of each component:

#	component	closed form
1	A very smooth function	symbolic regression (PySR) on the posterior mean, largest gap 0.68 % of the cor
2	A periodic function with a period of 7.0 days	Fourier series, 8 harmonics, largest gap 0.053 % of the component
3	A periodic function with a period of 6.0 months	Fourier series, 3 harmonics, largest gap 0.0033 % of the component
4	Uncorrelated noise	none: the covariance is the formula

Search path and stopping criterion

The 2014 configuration searches a fixed number of levels and keeps the lowest BIC; it has no stopping rule. The table follows the winner of each level. A change of structure adds or removes terms; a level that keeps the structure only refits its parameters. BIC differences read on the scale of Kass and Raftery (1995): under 2 barely worth mentioning, 2 to 6 positive, 6 to 10 strong, over 10 very strong.

level	BIC	change	structure	components	closed	params
1	10266.6		SE + WN	2	0	3
2	10208.0	-58.6	Per + SE + WN	3	1	6
3	9703.4	-504.6	Per x Per + SE + WN	3	0	8
4	9400.6	-302.8	Per + SE + WN	3	0	6
5 (kept)	9398.3	-2.3	Per + Per + SE + WN	4	2	9
6	9398.3	+0.0	Per + Per + SE + WN	4	2	9
7	9398.3	+0.0	Per + Per + SE + WN	4	2	9
8	9398.3	+0.0	Per + Per + SE + WN	4	2	9
9	9398.3	+0.0	Per + Per + SE + WN	4	2	9
10	9398.3	+0.0	Per + Per + SE + WN	4	2	9



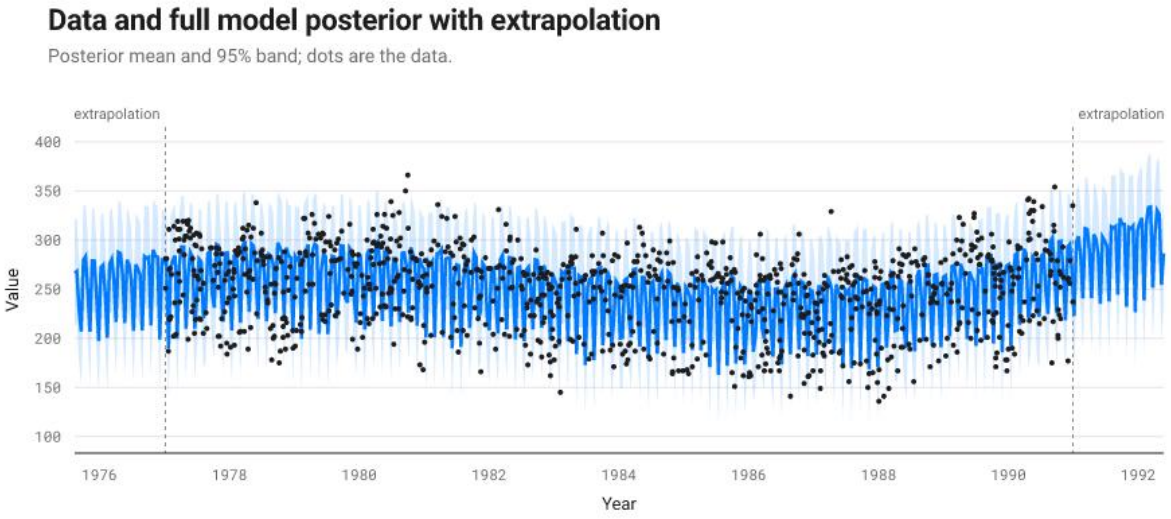
BIC per search level

Posterior weight of each structure met, exp(-BIC/2) normalised under a flat prior, after the Bayesian machine scientist (Guimerà et al., 2020): where the search keeps one structure, the weights say how much the data prefer it to the others. The 2 structure(s) under 0.001 are left out.

structure	lowest BIC	weight
Per + Per + SE + WN	9398.3	0.763
Per + SE + WN	9400.6	0.237

- The search ran 10 levels (the published 2014 search: ten levels, no stopping rule) and keeps the model of level 5, Per + Per + SE + WN, BIC 9398.3.
- The best different structure, Per + SE + WN (level 4), is 2.3 above it: positive evidence for the kept one on the Kass and Raftery scale.

- Weighed by exp(-BIC/2) under a flat prior, as the Bayesian machine scientist weighs expressions, the structures met give the kept one a posterior weight of 0.763 and the next, Per + SE + WN, 0.237: the data favour the kept one without ruling the next out.
- BIC stopped falling at level 5. The 5 level(s) searched past it kept its structure and lowered BIC by 0.00 in all, less than the tolerance of 0.1: going on buys nothing the data pay for.
- A rule stopping after 1 level(s) without a fall above 0.1 would have ended the search at level 6 and kept the same model.
- A rule stopping after 2 level(s) without a fall above 0.1 would have ended the search at level 7 and kept the same model.
- The kept expression has 9 parameters for 1000 points: its BIC is a fit term 2 NLL = 9336.1 plus a complexity term p ln n = 62.2, each extra parameter costing 6.91; it splits into 4 components, 2 of them with a closed form.



Data and full posterior

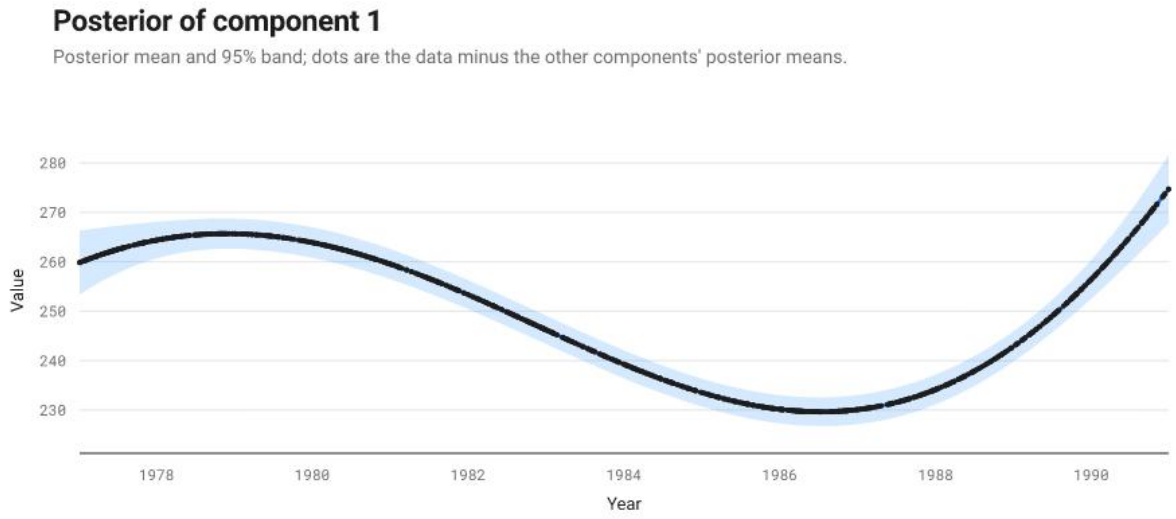
Detailed discussion of the components

Component 1: A very smooth function

This component is a very smooth function.

This component explains 9.7% of the data's variance on its own; with the components above it, 9.7%. Adding it changes the cross-validated MAE by -86.5%.

Extrapolation. This component is assumed to continue very smoothly but is also assumed to be stationary so its distribution will eventually return to the prior. The prior distribution places mass on smooth functions with a marginal mean of zero and a typical lengthscale of 10.9 years.



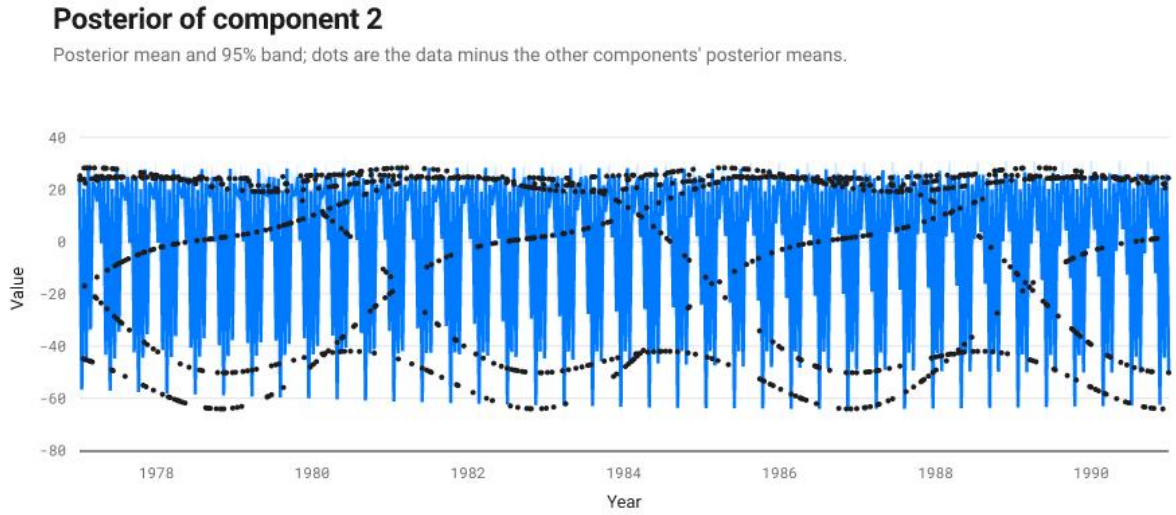
Component 1

Component 2: A periodic function with a period of 7.0 days

This component is periodic with a period of 7.0 days. The shape of this function within each period has a typical lengthscale of 25.1 hours.

This component explains 56.3% of the data's variance on its own; with the components above it, 67.1%. Adding it changes the cross-validated MAE by -43.1%.

Extrapolation. This component is assumed to continue periodically with a period of 7.0 days.



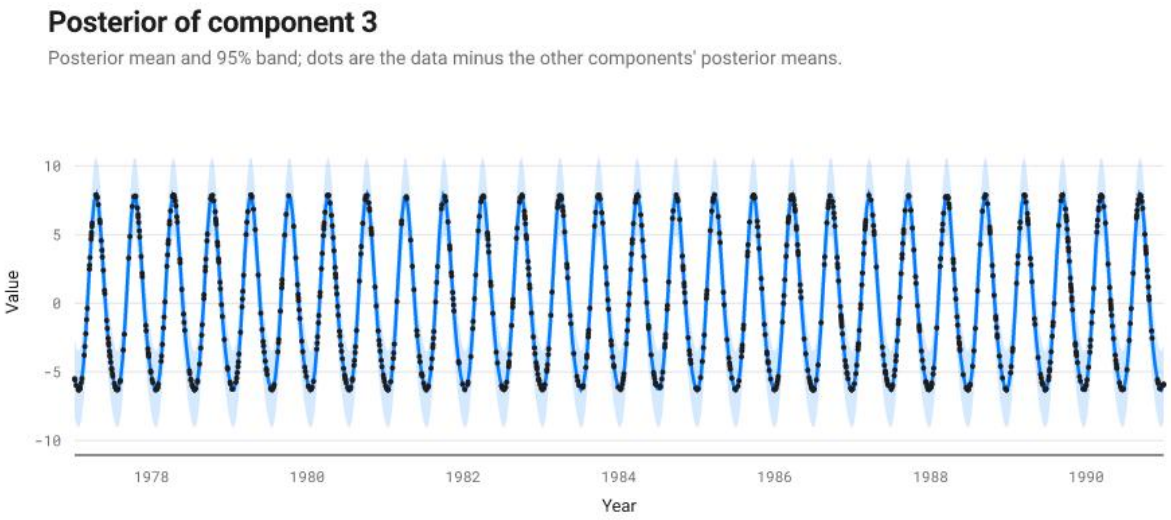
Component 2

Component 3: A periodic function with a period of 6.0 months

This component is periodic with a period of 6.0 months. The shape of this function within each period has a typical lengthscale of 2.3 months.

This component explains 1.3% of the data's variance on its own; with the components above it, 68.7%. Adding it changes the cross-validated MAE by -2.1%.

Extrapolation. This component is assumed to continue periodically with a period of 6.0 months.



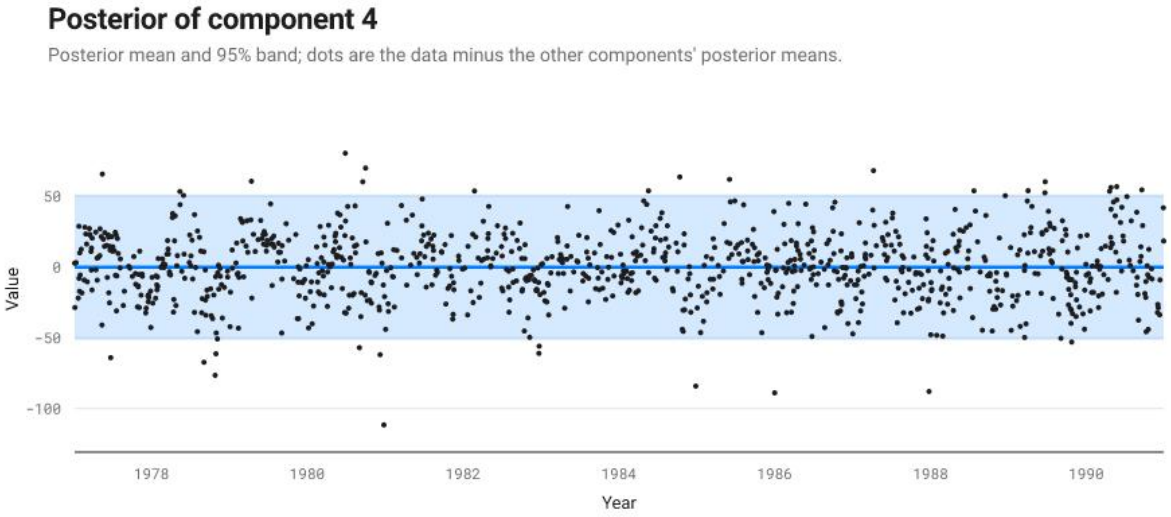
Component 3

Component 4: Uncorrelated noise

This component models uncorrelated noise.

This component explains 32.3% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component assumes the uncorrelated noise will continue indefinitely.



Component 4

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.144	0.568	**0.040**	0.124	0.618	0.674	**0.000**
2	**0.000**	**0.000**	0.718	0.306	**0.000**	0.756	**0.000**
3	0.994	0.708	0.862	0.784	0.906	0.936	**0.003**
4	0.842	**0.032**	0.986	0.992	0.990	0.920	**0.000**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.